

ePortfolio TM126: Activity 11.2.3 configure DHCP on a Wireless Router

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Week : 3

Introduction:

This activity demonstrates how to implement a DHCP configuration. The objective is for the client to connect to a wireless router via the DHCP server, which must assign an automatic IP address and verify network connectivity.

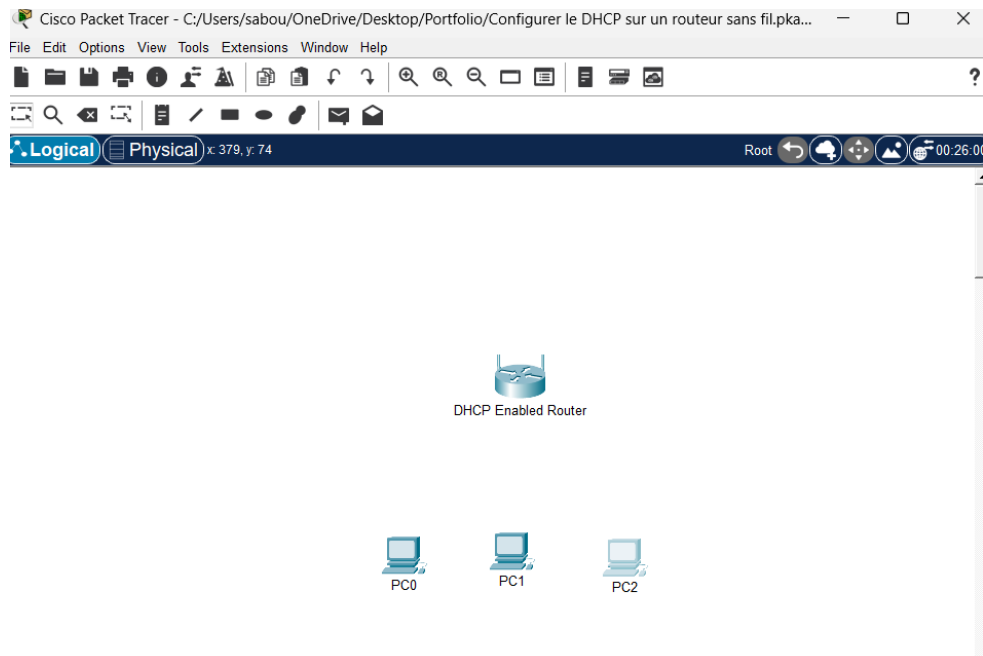
Objectives:

1. Connect three computers to a wireless router.
2. Set the DHCP parameter to a specific network range.
3. Configure clients to obtain their address via DHCP.

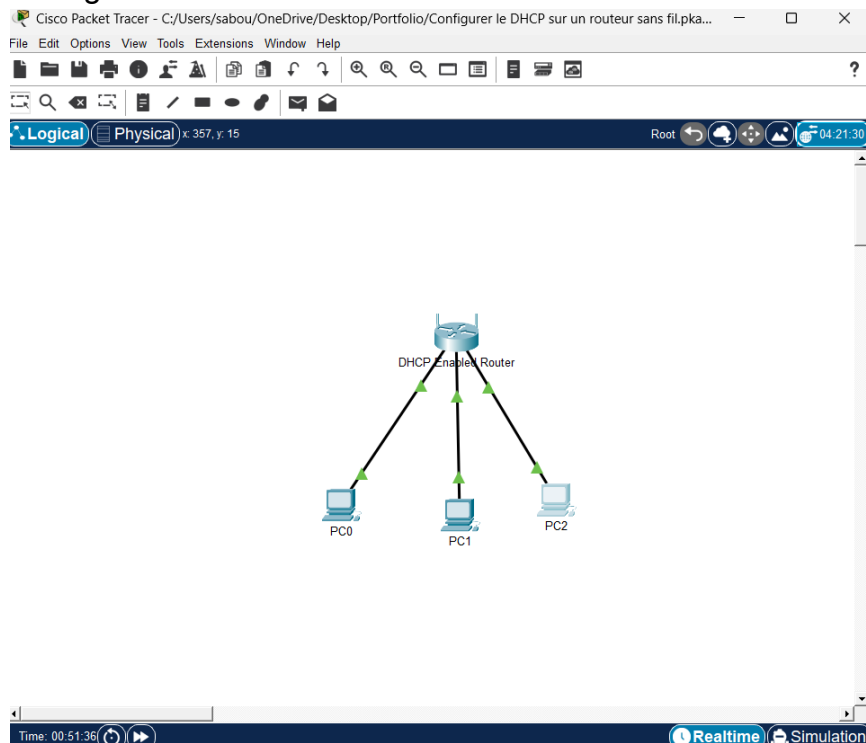
Instructions

Part 1: Set up the network topology

- a) Add three generic PCs.



- b) Connect each PC to an Ethernet port to the wireless router using straight-through cables.



Part 2: Observe the default DHCP settings

- a) Verification of IP addresses obtained automatically on PC0 via the DHCP server

PC0 has obtained an automatic IP address.

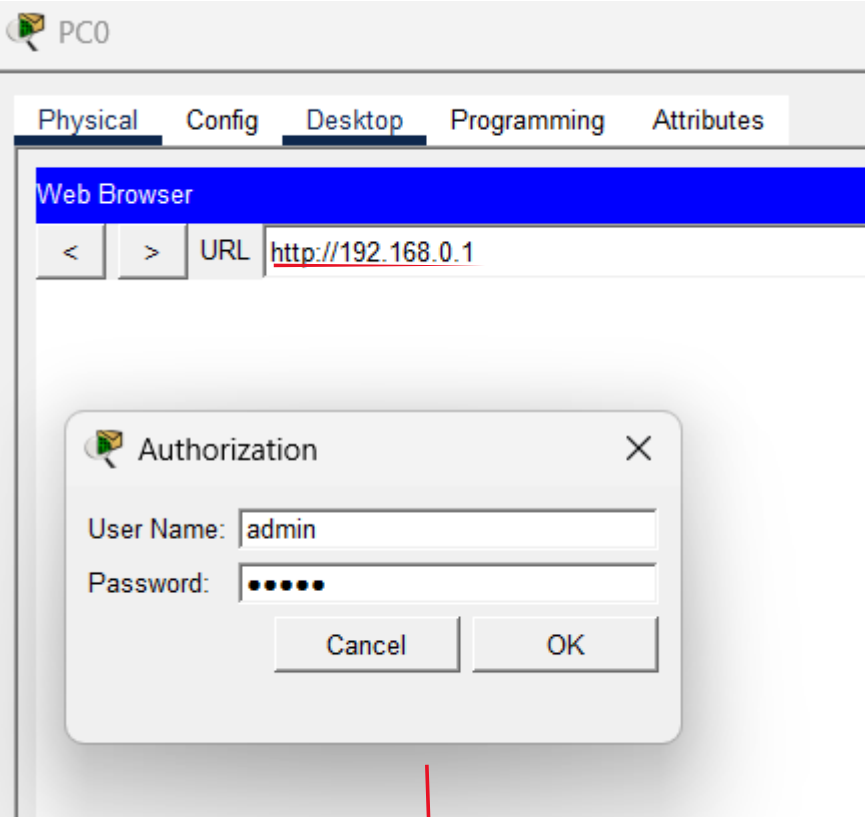
The image displays two side-by-side screenshots of the PC0 configuration window in Cisco Packet Tracer. Both screenshots show the "Desktop" tab with the "IP Configuration" section expanded. The left screenshot shows the "Static" tab selected, with the "Static" radio button highlighted. The right screenshot shows the "DHCP" tab selected, with the "DHCP" radio button highlighted. A red arrow points from the "Static" tab in the left window to the "DHCP" tab in the right window. The right window shows the automatically assigned IP address 192.168.0.100, Subnet Mask 255.255.255.0, Default Gateway 192.168.0.1, and DNS Server 0.0.0.0.

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	
Subnet Mask	
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

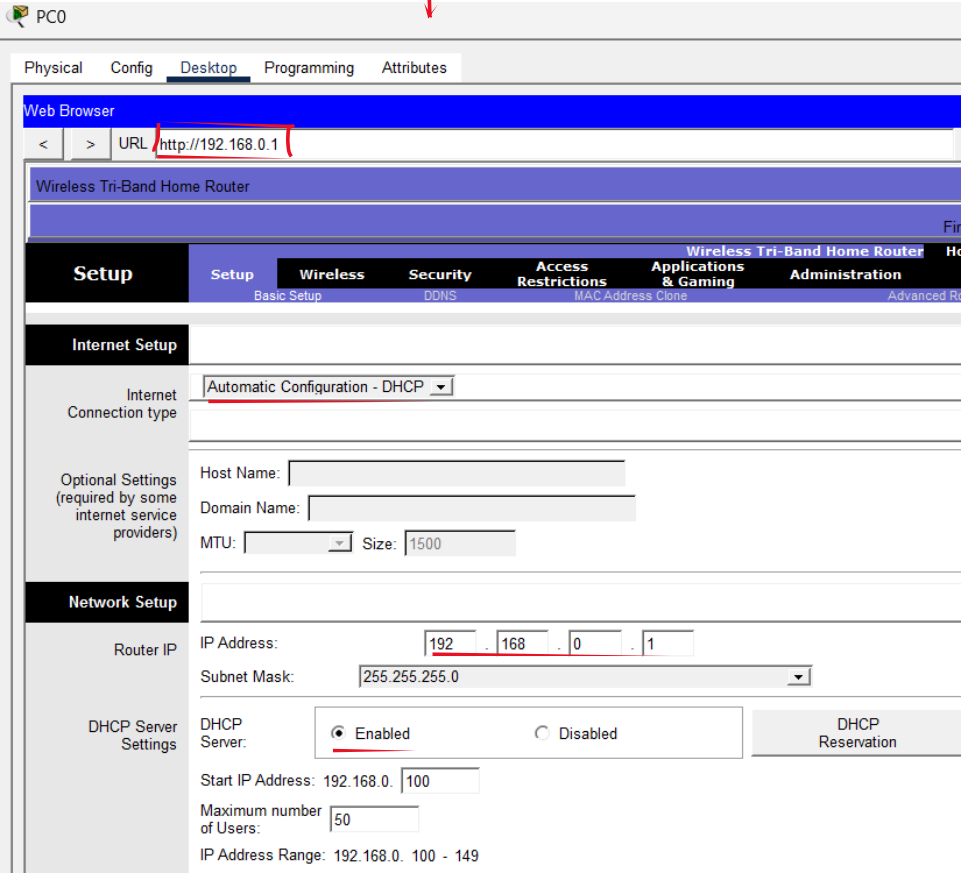
Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.0.100
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1
DNS Server	0.0.0.0

The IP address of the default gateway: is **192.168.0.1**

- b) Access the router's administration panel check The DHCP server and its setting



The top screenshot shows a PC0 interface with a 'Web Browser' window. The URL bar contains 'http://192.168.0.1'. An 'Authorization' dialog box is open in the foreground, prompting for 'User Name' (admin) and 'Password' (masked with dots). A red arrow points from the 'OK' button in the dialog to the bottom screenshot.



The bottom screenshot shows the router's administration panel. The 'Setup' tab is selected, and the 'Internet Setup' section is expanded. The 'Internet Connection type' is set to 'Automatic Configuration - DHCP'. The 'Router IP' section shows the IP Address as 192.168.0.1 and the Subnet Mask as 255.255.255.0. The 'DHCP Server Settings' section shows the DHCP Server is 'Enabled', the Start IP Address is 192.168.0.100, the Maximum number of Users is 50, and the IP Address Range is 192.168.0.100 - 149.

Part 3: Change the default IP address of the wireless router.

3a) Changed the router IP address from 192.168.0.1 to 192.168.5.1

The image shows a web browser window with the URL `http://192.168.5.1`. An "Authorization" dialog box is displayed in the center, asking for a "User Name:" and "Password:" with "Cancel" and "OK" buttons. Below the dialog, a red arrow points to the "Config" tab in the browser's navigation bar. The configuration page is open, showing the "Internet Setup" and "Network Setup" sections. In the "Network Setup" section, the "Router IP" is set to `192.168.5.1`, which is highlighted with a red box. The "Subnet Mask" is `255.255.255.0`. The "DHCP Server" is set to "Enabled". The "Start IP Address" is `192.168.5.26`, the "Maximum number of Users" is `75`, and the "IP Address Range" is `192.168.5.26 - 100`. The "Client Lease Time" is `0` minutes. There are also fields for "Static DNS 1", "Static DNS 2", "Static DNS 3", and "WINS".

Web Browser

URL `http://192.168.5.1` Go Stop

Authorization

User Name: Password: Cancel OK

Physical Config Desktop Programming Attributes

Web Browser

URL `http://192.168.5.1` Go Stop

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers)

Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP

IP Address: 192 . 168 . 5 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings

DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.5.26

Maximum number of Users: 75

IP Address Range: 192.168.5.26 - 100

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0

Static DNS 2: 0 . 0 . 0 . 0

Static DNS 3: 0 . 0 . 0 . 0

WINS: 0 . 0 . 0 . 0

Part 4: Change the default DHCP range of addresses.

Defining an IP address range from 192.168.5.100 to **192.168.5.126** for the DHCP server and Maximum Number of Users to **75**.

Physical Config **Desktop** Programming Attributes

Web Browser URL: http://192.168.5.1 Go Stop

Wireless Tri-Band Home Router

Setup Setup Wireless Security Access Restrictions Applications & Gaming Administration

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers): Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP: IP Address: 192 . 168 . 5 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings: DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.5. 100

Maximum number of Users: 50

IP Address Range: 192.168.5. 100 - 149

Client Lease Time: 0 minutes (0 means one day)

Static DNS 1: 0 . 0 . 0 . 0

Physical Config **Desktop** Programming Attributes

Web Browser URL: http://192.168.5.1 Go Stop

Wireless Tri-Band Home Router Firmware Version: v0.9.7

Setup Setup Wireless Security Access Restrictions Applications & Gaming Administration HomeRouter-PT-AC Status

Internet Setup

Internet Connection type: Automatic Configuration - DHCP

Optional Settings (required by some internet service providers): Host Name: Domain Name: MTU: Size: 1500

Network Setup

Router IP: IP Address: 192 . 168 . 5 . 1 Subnet Mask: 255.255.255.0

DHCP Server Settings: DHCP Server: ☒ Enabled ☐ Disabled DHCP Reservation

Start IP Address: 192.168.5. 126

Maximum number of Users: 75

IP Address Range: 192.168.5. 126 - 200

Record the IP address for PC0: the IP address for PC0 is **192.168.1.126**

```
Command Prompt

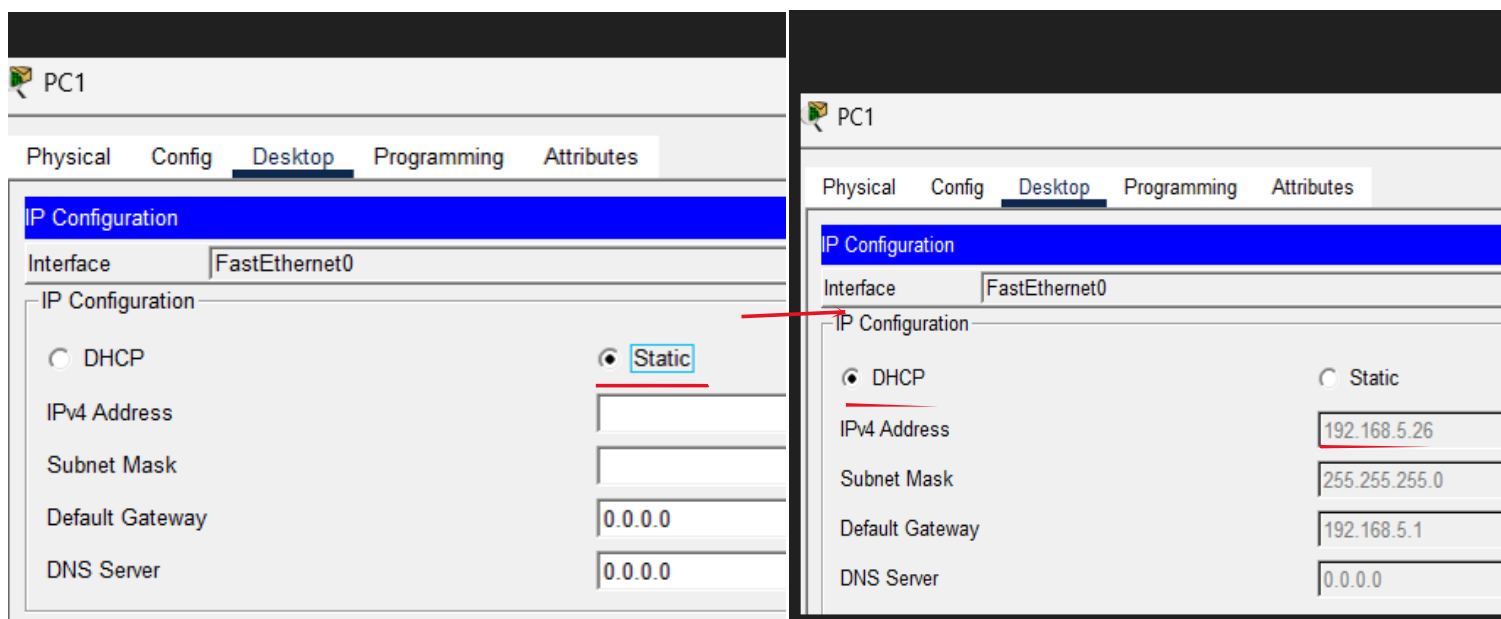
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::2D0:BCFF:FE68:6DE7
IPv6 Address.....: ::
IPv4 Address.....: 192.168.5.126
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.5.1
```

Part 5: Enable DHCP on the other PC1

Testing the automatic acquisition of new IP addresses on PC1



Record the IP address for PC1: the IP address for PC1 is **192.168.1.127**

PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::202:4AFF:FE7A:9264
IPv6 Address.....: ::
IPv4 Address.....: 192.168.5.127
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.5.1
```

Enable DHCP on **PC2**

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ **Static**

IPv4 Address

Subnet Mask

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☒ **DHCP** ☐ Static

IPv4 Address 192.168.5.128

Subnet Mask 255.255.255.0

Default Gateway 192.168.5.1

DNS Server 0.0.0.0

C:\>ipconfig

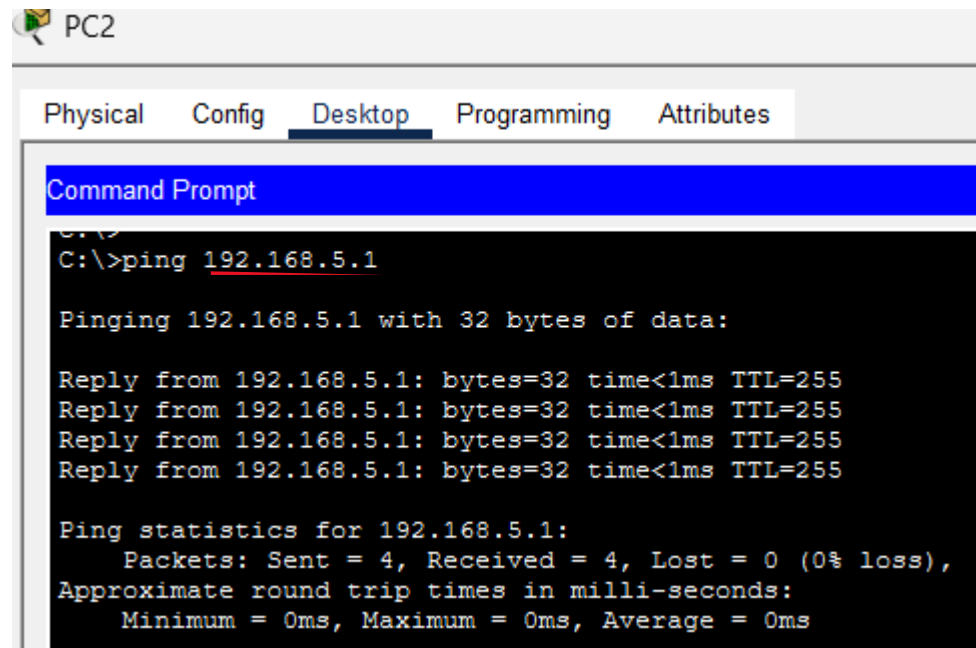
```
FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::210:11FF:FEA6:C864
IPv6 Address.....: ::
IPv4 Address.....: 192.168.5.128
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
                        192.168.5.1
```

Part 6: Verify Connectivity

Ping between machines to validate network communication.

From PC2 I pinged **192.168.5.1** to ping the wireless router.



The screenshot shows a Windows PC interface with a taskbar at the top. The 'Desktop' tab is selected in the top navigation bar. Below it, a 'Command Prompt' window is open, displaying the results of a ping command to 192.168.5.1. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The ping statistics indicate that all four packets were sent and received, with 0% loss, and the round trip times were all 0ms.

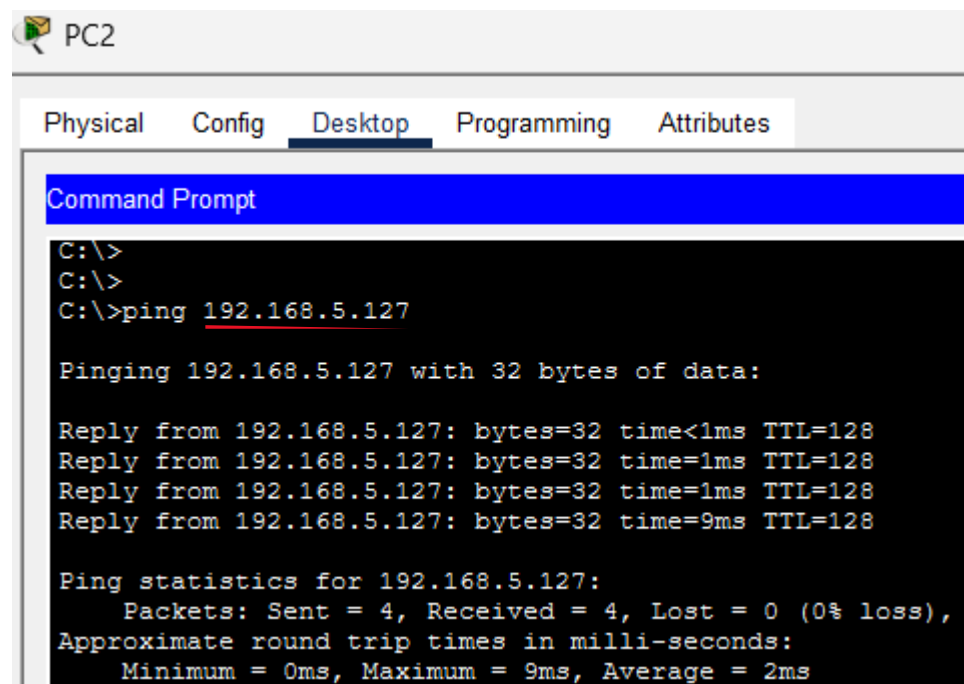
```
C:\>ping 192.168.5.1

Pinging 192.168.5.1 with 32 bytes of data:

Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255
Reply from 192.168.5.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.5.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

From PC2 I ping **192.168.5.127** to ping PC1



The screenshot shows a Windows PC interface with a taskbar at the top. The 'Desktop' tab is selected in the top navigation bar. Below it, a 'Command Prompt' window is open, displaying the results of a ping command to 192.168.5.127. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 128. The ping statistics indicate that all four packets were sent and received, with 0% loss, and the round trip times were all 0ms.

```
C:\>
C:\>
C:\>ping 192.168.5.127

Pinging 192.168.5.127 with 32 bytes of data:

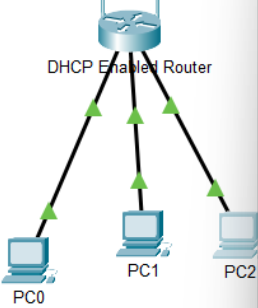
Reply from 192.168.5.127: bytes=32 time<1ms TTL=128
Reply from 192.168.5.127: bytes=32 time=1ms TTL=128
Reply from 192.168.5.127: bytes=32 time=1ms TTL=128
Reply from 192.168.5.127: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.5.127:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 2ms
```


Cisco Packet Tracer - C:\Users\sabou\Downloads\P1834897_Configure DHCP on a Wireless Router.pka - Guest - 2025-11-26 21:08:23

File Edit Options View Tools Extensions Window Help

Logical Physical x 700, y: 706



PT Activity: 00:09:33

Part 5: Enable DHCP on the other PCs.

- Click PC1.
- Select Desktop tab.
- Select IP Configuration.
- Click DHCP.

Record the IP address for PC1:

- Close the configuration window.
- Enable DHCP on PC2 following the steps for PC1.

Part 6: Verify connectivity

- Click PC2 and select the Desktop tab.
- Select Command Prompt.
- Enter `ipconfig` at the prompt to view the IP configuration.
- At the prompt, enter `ping 192.168.5.1` to ping the wireless router.
- Enter `ping 192.168.5.126` to ping PC0 at the prompt.
- At the prompt, enter `ping 192.168.5.127` to ping PC1.
- The pings to all devices should be successful.

Time Elapsed: 00:09:33 Completion: 100%

☐ Top ☐ Dock [Check Results](#) [Back](#) 1/1 [Next](#)

Cisco Packet Tracer - C:\Users\sabou\Downloads\P1834897_Configure DHCP on a Wireless Rou... - Guest - 2025-11-26 21:10:02

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:10:02

Congratulations Guest! You completed the activity.

Overall Feedback [Assessment Items](#) Connectivity Tests

[Expand/Collapse All](#) [Show Incorrect Items](#)

Assessment Items	Status	Points	Comp
Network			
DHCP Enabled Router			
DHCP Server			
Pools			
Pool linksysPool			
Default Gateway	Correct	1	Ip
Max User	Correct	1	Ip
Start IP address	Correct	1	Ip
PC0			
Ports			
FastEthernet0			
IP Address	Correct	1	Ip
Subnet Mask	Correct	1	Ip
PC1			
Ports			
FastEthernet0			
IP Address	Correct	1	Ip
Subnet Mask	Correct	1	Ip
PC2			
Ports			
FastEthernet0			
IP Address	Correct	1	Ip
Subnet Mask	Correct	1	Ip

Component	Items/Total	Score
Ip	9/9	9/9

Score : 9/9
Item Count : 9/9

Personal reflection

This activity allowed me to discover the role of the DHCP protocol. Thanks to this activity and the manipulations I performed, I understood how a DHCP server works. Configuring a wireless router also allowed me to strengthen my technical skills in all aspects of network management.